

Welcome to PH142 and the PPDAC framework

What is this class?

Statistics is Everywhere

PPDAC - the approach we
will use to answering
questions with statistics

PPDAC Example 1: A
smoking behaviour study

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Guess the year

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In this year, UC Berkeley established a statistics department (split from mathematics) and hired David Blackwell - the first African American to receive tenure at UC Berkeley, and the first African American elected to the National Academy of Science (10 years after his appointment)

Quote from Dr. Blackwell

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Basically, I'm not interested in doing research and I never have been... I'm interested in understanding, which is quite a different thing. And often to understand something, you have to work it out yourself because no one else has done it.

- ▶ quoted from a 2007 New York Times article

Today's Goals

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Welcome and orientation to the class

My goals for our time together

Course logistics

Talk about the framework we use in the class (PPDAC)

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Who am I?

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Our Teaching team

Lead GSI: Sian Louise Hong

Section GSIs:

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Lecture/Labs/Office Hours

Ed Discussion

Quizzes and exams

Data project

When in doubt - check the website and the Ed announcements

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Frequently asked questions so far

Do I have to attend lecture/section?

Do I need the textbook?

Do I need to know programming?

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How to get help with code

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- ▶ Ask questions during labs/discussion sections, office hours, or on Ed discussion forum. Use the appropriate thread!
- ▶ Develop your online search skills. For example if you have a `ggplot2` question, begin your google search with “r ggplot” and then describe your issues, e.g., “r ggplot how do I make separate lines by a second variable”.
- ▶ The most common links that will appear are:
 - ▶ <https://stackoverflow.com>: Crowd-sourced answers that have been up-voted. The top answer is often the best one.
 - ▶ <https://ggplot2.tidyverse.org/>: The official ggplot2 webpage is very helpful.
 - ▶ <https://community.rstudio.com/>: The RStudio community page.
 - ▶ <https://rpubs.com/>: Web pages made by R users that often contain helpful tutorials.

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Figure 1: Will I get an A?

a pre-emptive apology

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(credit to xkcd.com for the comic)

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Ongoing evolution of the course

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From Derivation to hands on programming

Co-Development of course with Dr. Riddell

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In this class, we are going to think about

- ▶ **DATA** - how we gather, display, and summarize information
- ▶ **Probability** - the role of chance
- ▶ **Statistics** - the science of drawing statistical conclusions from data using a knowledge of probability

My goals for you

Foundational concepts in probability and biostatistics

How to answer questions with data:

- ▶ your ability to critically assess statistical information presented to you in scientific and non-scientific fora
- ▶ your sense of how to approach answering real world questions with data
- ▶ develop your statistical intuition around variability and chance
- ▶ develop your toolkit for visualization, summarizing and testing simple relationships
- ▶ your ability to concisely and accurately describe statistical methods and results

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Three parts

- ▶ **Part I:** learning to explore and summarize univariate and bivariate distributions
- ▶ **Part II:** classical problems in probability, some commonly used probability distributions, and the central limit theorem
- ▶ **Part III:** statistical inference, making inference about populations using statistics constructed from samples

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This is not a math class

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Statistics is often classified as a branch of math, but I'd argue that it is more important to **focus on the connections that statistics has with science** (how we can learn about the world through data)

Though it is true that statistics uses math (and sometimes fairly advanced math!), **not much math is needed** to learn introductory statistics

In this class we will try, as much as possible, to **emphasize concepts** and help you develop your statistical intuition

This is not a programming class

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Statistics is often viewed as “just computer programming,” but this is an incorrect and dangerous characterization: **computer programming is simply a tool for conducting statistical analysis**

The use of computer programming in statistics is—and should be—**quite different** than approaches to non-statistical programming

We are using `r` programming in this course because it is an extremely useful skill, facilitates computation, and is desired in the job market

This is a relevant class

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I hope to convince everyone here that statistics is relevant to everyone

As is more and more apparent, public health statistics have relevance to important policy decisions

You also make many decisions during your day that are influenced by statistics

Statistics is not just relevant for public health, but also for other professions, including: education, journalism and law

As we'll try to illustrate via the recurring "statistics is everywhere" segments, statistics is useful for understanding the news and the world around us

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Statistics is Everywhere

Problems with scientific reporting



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Consequences of poor communication

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PPDAC - the approach we will use to answering questions with statistics

Problem

A clear statement of what we are trying to achieve.

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Three main problem types

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- ▶ **Descriptive**: learning about some particular attribute of a population
- ▶ **Causative/Etiologic**: do changes in an explanatory variable cause changes in a response variable?
- ▶ **Predictive**: how can we best predict the value of the response variable for an individual?

Problem type?

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- ▶ Insurance company: What is the probability (how likely is it) that a 25 year old unmarried male driver has a car accident?
- ▶ Health department: How many cases of influenza have we seen this season compared to last season?
- ▶ Health care system: If we treat patients with diabetes using medication X, will their insulin regulation be better or worse than medication y?

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The procedures we use to carry out the study.

- ▶ **Census** or **sample** from the target population?
 - ▶ How was the sampling conducted?
 - ▶ Was the sample random?
- ▶ Is the study prospective or retrospective?
- ▶ Is the study observational or experimental?

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The data which is collected according to the Plan.

- ▶ How many observations do we have?
- ▶ How reliable are the measures?

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The data is summarized and analysed to answer the questions posed by the Problem.

We use our knowledge about probabilities to assess the role of chance in our findings.

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Conclusions are drawn about what has been learned about answering the Problem.

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Problem: Suppose we wish to study the smoking behavior of California residents aged 14-20 years.

In particular, we are interested in the *prevalence* of current smoking by gender.

What type of problem is this?

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Plan: We need to first choose a time period, because we know that smoking behavior has changed immensely over time. It is unfeasible to gather these data for all residents in California who are 14-20 years old.

Instead we conduct a *random sample* of size n persons. We collect their: age, gender, and smoking status.

Note that we need to decide how large n should be, and how to obtain the random sample. The latter question is, in particular, very important if we want to ensure that our sample is representative of the population of interest. Time and money also constrain how the sample will be collected.

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Data: Suppose that a random sample of 200 persons aged 14-20 was selected, yielding these data:

Gender	Number of smokers	Number of non-smokers	Total
Teen girls and women	32	66	98
Teen boys and men	27	75	102
Total	59	141	200

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Analysis: The proportion of women in the sample who smoke is $32/98 = 33\%$.
The proportion of men in the sample who smoke is $27/102 = 26\%$.

We would also like some idea as to how close this estimate is likely to be from the actual proportion in the population.

If we selected a second random sample of the same size, we would likely estimate different proportions for men and women. We will learn how to estimate the precision of these estimates.

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Conclusion: 33% of girls and women aged 14-20 and 26% of boys and men of the same age group are current smokers in California in 2018 (plus a measure of uncertainty).

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Life expectancy is one of the core measures used in public health to comment on the well-being of groups of people. Differences in life expectancy by race/ethnicity, for individuals living in the same region can reflect underlying inequalities in policies, access to care, food environments, structural and systemic racism, among other potential causes.

Research objective (Problem)

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The purpose of this short report is to visualize life expectancy among black and white men and women in California between 1969 and 2013.

We are interested in whether there are differences by group and whether these differences have changed over time.

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Death certificates in the United States include race/ethnicity, age at death, and date of death and capture all deaths of US residents. These data are aggregated by the CDC's National Cancer Institute into the SEER*Stat software. Previously, Riddell et al.¹, analyzed these data to compute estimated trends in life expectancy for non-Hispanic black and white men and women, for 40 US states between 1969 and 2013. States without enough data were excluded from these analyses.

To carry out this short report, we will use data from Riddell et al. to visualize trends in life expectancy as part of an exploratory data analysis. In particular, we will plot time trends for black and white men and women in California.

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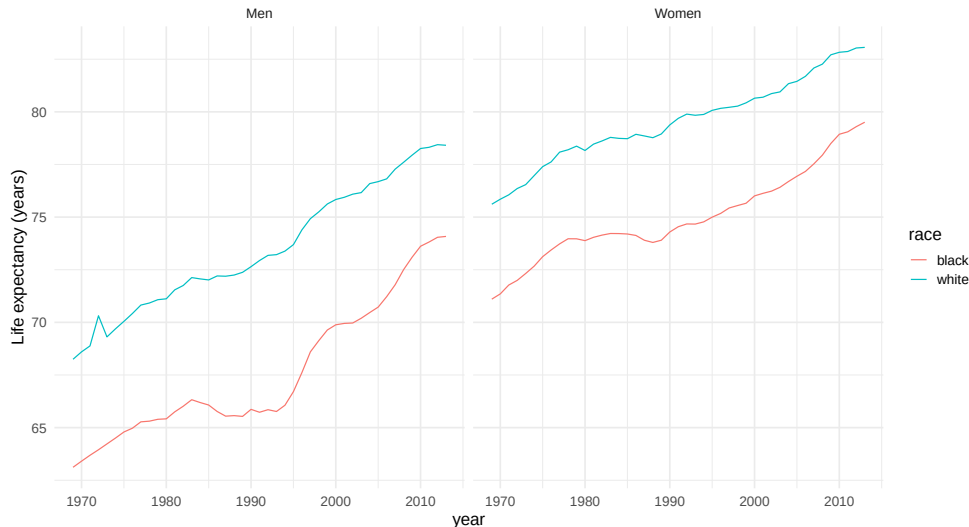
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Here are the first few rows of these data for California:

state	stabbrs	year	sex	Census_Region	Census_Division	LE	race
California	CA	1969	Female	West	Pacific	75.61137	white
California	CA	1969	Male	West	Pacific	68.24766	white
California	CA	1970	Female	West	Pacific	75.84916	white
California	CA	1970	Male	West	Pacific	68.59865	white
California	CA	1971	Female	West	Pacific	76.05663	white

Trends in life expectancy for black and white men and women in California



What is this class?

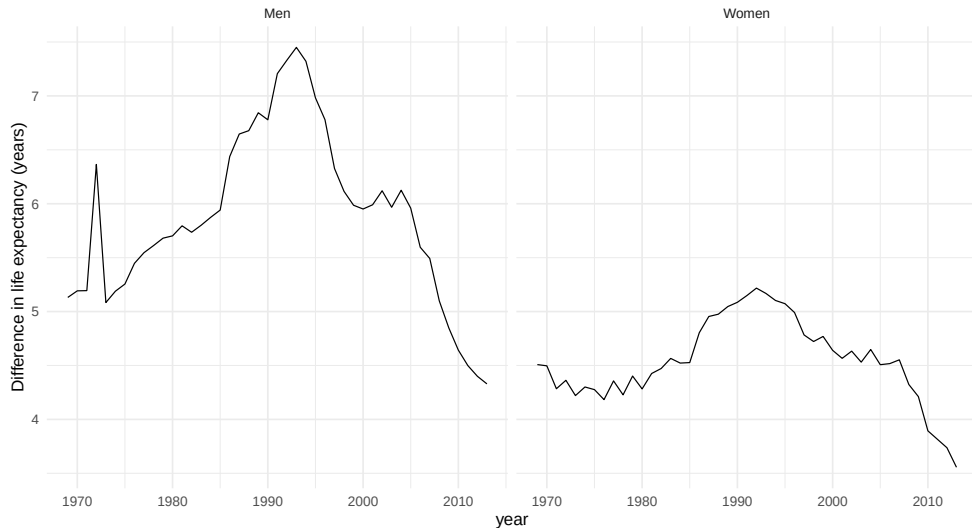
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Difference in life expectancy between black and white men and women in California



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The difference in life expectancy in 1969 between non-Hispanic blacks and whites was 5.1 years for men and 4.5 for women in California.

By 2013, the difference was 4.3 years for men and 3.6 for women in California.